

EcoMetric Thermal Systems Inc.

Environmental Product Declaration

[Product Image: EcoMetric 500RT Centrifugal Chiller]

DRAFT: NOT THIRD-
PARTY VERIFIED

EPD Program and Program Operator Name, Address, Logo, and Website	NSF Certification, LLC 789 North Dixboro Road, Ann Arbor, MI, 48105, United States https://www.nsf.org/
General Program Instructions and Version Number	Part A: Life Cycle Assessment Calculations and Report Requirements Version 4.0
Manufacturer Name and Address	EcoMetric Thermal Systems Inc. 13995 Pasteur Boulevard Palm Beach Gardens, Florida 33418
Declaration Number	EPD11017
Product and Functional Unit	EcoMetric 500RT Centrifugal Chiller, 1 ton chilling capacity
Reference PCR and Version Number	Part A: Life Cycle Assessment Calculation Rules and Report Requirements (UL Environment, V4.0, 2022) Part B: Water Cooled Chiller EPD Requirements (UL Environment V2.0, 2018)
Product's Intended Application and Use	The function of the chiller included within this study is to provide chilled water for use in cooling the interior of a building, for a functional unit of 1 ton chilling capacity.
Product RSL	25 years
Markets of Applicability	North America, Global
Date of Issue / Period of Validity	12/17/2024 – 5 Years from the date of issue
EPD Type	Product Specific
Range of Dataset Variability	N/A
EPD Scope	Cradle-to-Grave
Year of reported manufacturer primary data	2023
LCA Software and Version Number	EcoMetric EPD Platform v2.4 (One Click LCA / Ecoinvent compatible)
LCI Database and Version Number	Ecoinvent v3.12 (Cut-off system model)
LCIA Methodology and Version Number	TRACI 2.1 + CML-IA v4.1 + PEF / EN 15804+A2
The sub-category PCR review was conducted by:	Lise Laurin, EarthShift Global Sean Beilman, BCER Engineering, Inc. François Charron-Doucet, Group AGÉCO
This declaration was independently verified in accordance with ISO 14025: 2006. The PCR chosen conforms to ISO 21930:2017. ☐ INTERNAL ☐ EXTERNAL <i>(Green/Red Draft)</i>	<i>(Pending Verification)</i>
This life cycle assessment was conducted in accordance with ISO 14044 and the reference PCR by:	Shashikumar M S, HCLTech
This life cycle assessment was independently verified in accordance with ISO 14044 and the reference PCR by:	<i>(Pending Verification)</i>

Limitations:

Environmental declarations from different programs (ISO 14025) may not be comparable.

Comparison of the environmental performance of products using EPD information shall be based on the product's use and impacts at the building level, and therefore EPDs may not be used for comparability purposes when not considering the building use phase as instructed under this PCR. Previous versions of the PCR vary in the prescribed default refrigerant leakage rate and do not necessitate the reporting of B4 impacts. These variations make it imperative to ensure that EPDs are aligned with all the following conformance requirements for comparisons.

Full conformance with the PCR for Water Chillers allows EPD comparability only when all stages of a life cycle have been considered, when they comply with all referenced standards, use the same sub-category PCR, and use equivalent scenarios with respect to construction works. However, variations and deviations are possible. Example of variations: Different LCA software and background LCI datasets may lead to differences results for upstream or downstream of the life cycle stages declared.

No use phase grid mix is explicitly defined by the PCR. As such, Carrier has elected to report multiple use phase electricity demand scenarios for increased reader utility. This reporting is described in the text of the EPD.

1 – GENERAL INFORMATION

Description of Company

Carrier is the leading global provider of healthy, safe, and sustainable building and cold chain solutions with a world-class, diverse workforce. Through performance-driven culture, shareholder value is driven by growing earnings and investing strategically to strengthen its position in the market. Carrier's industry leading solutions and services are designed to reduce energy consumption and facility operating costs in HVAC & Refrigeration.

Product Description

This environmental product declaration covers the following products from CSI division 23 64 16.16: EcoMetric 500RT Centrifugal Chiller. Further explanatory material may be obtained through the company website: <https://www.carrier.com>.

The AquaEdge® 19DV is a water-cooled centrifugal chiller that utilizes a two-stage back-to-back compressor and an oil-free ceramic bearing system to deliver more operating range and consistent efficiency. Product shown in Figure 1.

EcoMetric 500RT Centrifugal Chiller has cooling capacities ranging from 350 - 1,150 tons. Further 500 ton which is the most common configuration by sales volume is assessed for the purpose of this EPD. The chiller models allow for a nearly endless amount of variability in configuration based on the specifications and needs of the ultimate consumer.

This configuration uses refrigerant, R134A.

No substances required to be reported as hazardous according to the US Resources Conservation and Recovery Act, Subtitle 3 are associated with the production of this product.

Technical Data

The EcoMetric 500RT Centrifugal Chiller series is certified to the following standards:

- High tier variable speed starter equipped with harmonic filter (optional), total harmonic distortion (THD) ≤5% and fully complies with IEEE519 standard.
- AquaEdge® 19DV chillers can achieve up to 7.3 (0.4818 kW/Ton) full load COPR and 12.3 (0.2859 kW/Ton) IPLV.IP at AHRI conditions.
- AquaEdge® 19DV chillers can meet 18001 standards recommended by Occupational Health and Safety Advisory Services (OHSAS).
- ASME Section VIII Div. 1 "U" stamped certified.
- Certified in accordance with the AHRI Water-Cooled Water-Chilling and Heat Pump Water-Heating Packages Certification Program (AHRI Standard 550/590).
- Certified units may be found in the AHRI Directory at <http://www.ahridirectory.org>.

Table 1 shows the technical specifications of the products, including any testing data as appropriate.

Table 1: Technical Data

Name	Value	Unit
Chilling Capacity	500	tons of refrigeration (RT)
Energy Efficiency* (100% Load at 85°F)	0.4818	kW/ton at AHRI 550/590 conditions
Energy Efficiency* (75% Load at 75°F)	0.3373	kW/ton
Energy Efficiency* (50% Load at 65°F)	0.2602	kW/ton
Energy Efficiency* (25% Load at 65°F)	0.3132	kW/ton

*Values from AHRI 550/590 Standard

The dimensions of each configuration as delivered to the customer are reported in Table 2.

Table 2: Product Dimensions

Dimension	500 ton	Unit
Length	5.2	m
Width	2.6	m
Height	3.1	m

Application

The function of the chiller included within this study is to provide chilled water for use in cooling the interior of a building, for a functional unit of 1 ton chilling capacity. In this study, the reference service life (RSL) of the products is 25 years. Therefore, after initial installation in a building with an estimated service life (ESL) of 75 years there will be 2 replacements needed.

Material Composition

The raw materials for the product were obtained from verified suppliers. The general compositions of the products are represented in Table 3.

Table 3: Material Composition per Functional Unit

Materials	500 ton
R134A Charged Chiller: Material Composition per Functional Unit	
Steel [kg]	9.46
Iron [kg]	0.61
Copper [kg]	2.60
Aluminum [kg]	0.00
Refrigerant [kg]	0.09

Materials		500 ton
Other [kg]		2.88
R134A Charged Chiller: Contribution to Total Material Composition		
Steel		60.49%
Iron		3.90%
Copper		16.62%
Aluminum		0.00%
Refrigerant		0.58%
Other		18.41%

Manufacturing

All major components are manufactured within ISO 9001/14001 facilities. The final assembly occurs at the main manufacturing facility. Components include the tube sheets, the evaporator, the condenser, and the control box.

Manufacture impacts of capital goods and infrastructure flows are excluded as they are estimated to impact, at most, 0.12 kg CO₂e per ton of chilling capacity over the life cycle of the manufacturing infrastructure, which is considered negligible. This assumption is in alignment with typical building product LCA practice.

The functional unit according to the PCR is 1 ton chilling capacity. The products under study have a reference service life (RSL) of 25 years. Table 4 shows additional details related to the functional unit.

Table 4: Functional Unit Details

Parameter	500 ton	Unit
Functional Unit	1 ton chilling capacity	-
Mass of one delivered product	7,775	kg
Conversion factor (kg per Functional Unit)	15.6	kg/ton of chilling capacity

Transportation

The product is delivered to the customer via truck. Transportation distances are set by the PCR and are shown in Table 5, along with additional transportation data. It is assumed that each chiller is shipped individually using a flatbed truck. Additionally, it is assumed that the return route for the truck is empty.

Table 5: Transport to Building Site (A4) per Functional Unit

Parameter	500 ton
Vehicle Type	>32000 Kg payload Flatbed Truck
Product Weight (Kg)	7,775.0
Fuel Efficiency [L/100km]	36.3
Fuel Type	Diesel
Distance [km]	750
Capacity Utilization [%]	24%
Gross density of products transported [kg/m ³]	369
Capacity utilization volume factor	<1

Installation

A crane is necessary to unload the chiller from the truck. It is assumed that the crane is operational for 3 hours and consumes diesel fuel based on crane manufacturer specs. Installation details are summarized in Table 6.

Table 6: Installation Scenario Details per Functional Unit

Parameter	500 ton	Unit
Other energy carriers by type – Diesel Fuel	2.71 MJ	MJ
Packaging Waste to Landfill	0.050	kg
Packaging Waste to Recycling	0.040	kg

RSL and ESL Scenarios

Standard interior building operating conditions are assumed throughout the use phase of the chiller. Details are provided in Table 7.

Table 7: Reference Service Life and Estimated Service Life of Chillers

Parameter	500 ton	Unit
Product RSL (years)	25	years
Building ESL (years)	75	years
Replacement Cycle(s)	2	number
Use Conditions (based on R134A refrigerant)	638,000	Annual energy demand (kWh)

Modules B2-B5 Scenarios

Impacts from refrigerant loss are assigned to the maintenance impacts module (B2). A 2% refrigeration loss and replacement rate is assumed based on guidance from the PCR.

Table 8: Maintenance (Module B2)

Parameter	500 ton	Unit
Maintenance process information	Replacement of leaked refrigerant assuming a 2% annual loss rate	n/a
Maintenance cycle (per RSL)	25 cycles	cycles
Maintenance cycle (per ESL)	75 cycles	cycles
Refrigerant replacement, R134A	0.04	kg
Direct emissions to ambient air, soil and water (refrigerant to air)	0.04	kg

Table 9: Replacement (Module B4)

Parameter	500 ton	Unit
Replacement cycles (ESL/RSL-1)	2 cycles	cycles
Direct emissions to ambient air, soil and water (refrigerant to air)	0.0	kg
Further assumptions for scenario development	90% of refrigerant is recovered at EOL but 10% is emitted to ambient air	As appropriate

Operational Energy Use (Module B6)

Water cooled chillers transfer heat away from spaces that require climate control by cooling water and allowing it to absorb heat from an environment. The operational efficiencies and annual energy demand are reported in Table 10.

Table 10: Operational Energy Efficiencies and Use (B6)

Name	Value	Unit
Chilling Capacity	500	tons of refrigeration (RT)
Electricity Consumption/Year (1 year)	638,000	kWh
Electricity Consumption/Year for ESL (75 Years)	47,850,000	kWh
Electricity Consumption/Year (per functional unit)	1,276.000	kWh/ton

Table 11: Operational Energy Use Phase

Country	Percent Contribution [%]	Emission Factor KgCO ₂ e/kWh
US	100	0.45

Disposal (Module C1-C4)

The chiller, in its entirety, is assumed to be delivered via truck to a recycling facility for dismantling and further product disposal. After separation at the facility, all metal components are assumed to be recycled and all plastic components are assumed to be landfilled. Table 12 shows the parameters for the end-of-life scenario.

Table 12: End-of-Life Scenario Details (C1-C4)

Parameter	500 ton
Collected separately	0
Collected as mixed construction waste [kg]	7,775.00
Waste to Reuse [kg]	0
Distance to Reuse [km]	0
Waste to Landfill [kg]	762.0
Distance to Landfill [km]	100
Waste to Incineration [kg]	0
Distance to Incineration [km]	0
Waste to Recycling [kg]	7,013.05
Distance to Recycling [km]	100

2 – LIFE CYCLE ASSESSMENT METHODOLOGY

This LCA is a Cradle-to-Grave study. An overview of the system boundary is shown in Figure 2 and a summary of the life cycle stages included in this LCA is presented in Table 13.

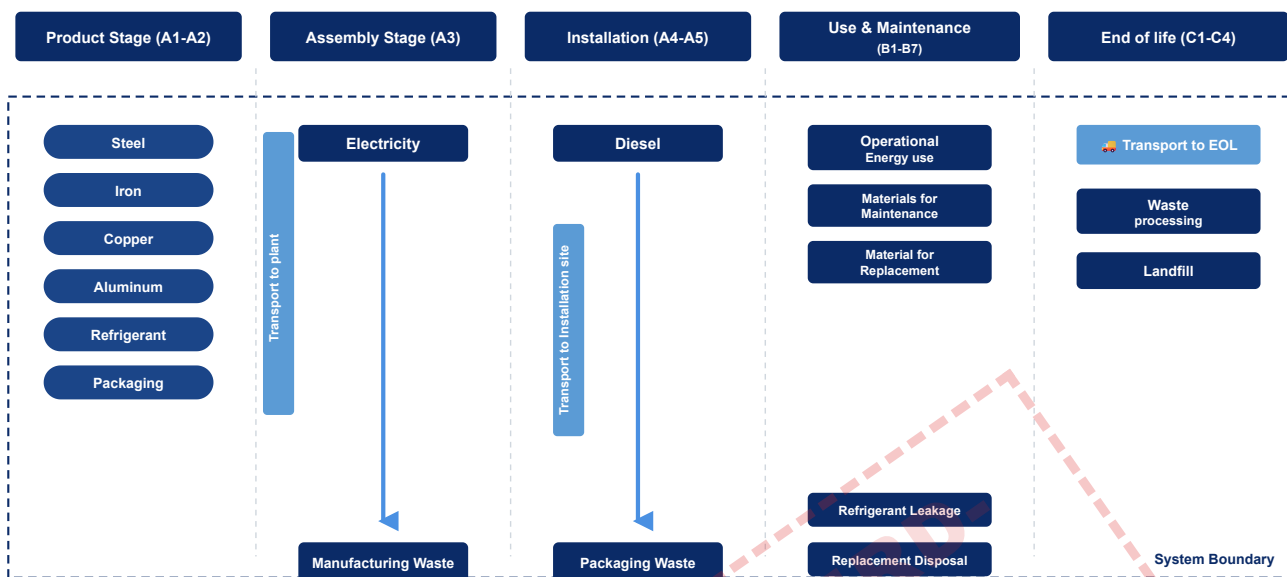


Figure 2: System Boundary Diagram

Table 13: Life Cycle Stages Included in the Study

Product stage			Assembly stage			Use stage							End of life stage				Beyond system boundaries		
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D	D	D	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	MND	MND	MND	
Raw materials	Transport	Manufacturing	Transport	Assembly	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Demolition	Transport	Waste processing	Disposal	Reuse	Recovery	Recycling	
US & Global	Global	US / China	Global	Global	Global	Global	-	Global	-	US & Global	-	-	Global	Global	Global	-	-	-	

X = Module Included in LCA, MND = Module not Declared

Cut-off Criteria

The study does not exclude any modules or processes which are stated mandatory in the reference standard and the applied Product Category Rule. The study does not exclude any hazardous materials or substances. The study includes all major raw material and energy consumption. All inputs and outputs of the unit processes, for which data is available for, are included in the calculation. There is no neglected unit process more than 1% of total mass or energy flows. The module specific total neglected input and output flows also do not exceed 5% of energy usage or mass.

The list of excluded materials and energy inputs include:

- As the tools used during the installation of the product are multi-use tools and can be reused after each installation, the per-declared unit impacts are considered negligible and therefore are not included.
- While required by the PCR, inclusion of capital goods and infrastructure flows is typical within LCA practice following construction product PCRs. Infrastructure flows for production facilities are assumed to be negligible (at most 0.12 kg CO₂e per ton).
- Some material inputs may have been excluded within datasets used; all conform to exclusion requirements of the PCR.

Allocation

General principles of allocation were based on ISO 14040/44. There are no products other than the product under study that are produced as part of the manufacturing processes. Since there are no co-products, no allocation based on co-products is required.

Throughout the study recycled materials were accounted for via the cut-off method. Under this method, impacts and benefits associated with the previous life of a raw material from recycled stock are excluded from the system boundary. Reprocessing and preparation impacts of recycled materials in the BOM are included.

Data Quality

Primary data was collected from the manufacturer specific to the temporal and technological aspects of the chiller life cycle. All primary data is from calendar year 2023. Secondary data was sourced from Ecoinvent Datasets, taking into consideration the most accurate and representative options. The data included is considered complete, consistent, and reproducible.

3 – LIFE CYCLE ASSESSMENT RESULTS

All results are given per functional unit, which is 1 ton chilling capacity. Each product under study is reported separately by life cycle stage. Characterization factors PEF / EN 15804+A2, CML-IA version 4.1, and TRACI 2.1 have been used throughout the study. Parameters used in the study are presented in Table 14, Table 15, Table 16, Table 17.

Table 14: Core Environmental Impact Parameters (EN 15804+A2, PEF)

Impact category / Parameter	Parameter Code	Unit	Characterization Model / Standard
Global warming potential total	GWP-total	kg CO ₂ eq.	Baseline model of 100 years of the IPCC (based on IPCC 2013)
Global warming potential fossil fuels	GWP-fossil	kg CO ₂ eq.	Baseline model of 100 years of the IPCC (based on IPCC 2013)
Global warming potential biogenic	GWP-biogenic	kg CO ₂ eq.	Baseline model of 100 years of the IPCC (based on IPCC 2013)
Global warming potential land use and land use change	GWP-luluc	kg CO ₂ eq.	Baseline model of 100 years of the IPCC (based on IPCC 2013)
Depletion potential of the stratospheric ozone layer	ODP	kg CFC-11 eq.	Scenarios of WMO 2014
Acidification potential, Accumulated Exceedance	AP	mol H ⁺ eq.	Seppälä et al. 2006, Posch et al. 2008
Eutrophication potential, freshwater	EP-freshwater	kg P eq.	EUTREND model, Struijs et al. 2009
Eutrophication potential, marine	EP-marine	kg N eq.	EUTREND model, Struijs et al. 2009
Eutrophication potential, terrestrial	EP-terrestrial	mol N eq.	Seppälä et al. 2006, Posch et al. 2008
Formation potential of tropospheric ozone	POCP	kg NMVOC eq.	Van Zelm et al. 2008
Abiotic depletion potential for non-fossil resources	ADP-minerals&metals	kg Sb eq.	CML 2002, Guinée et al. 2002
Abiotic depletion potential for fossil resources	ADP-fossil	MJ	CML 2002, Guinée et al. 2002
Water (user) deprivation potential	WDP	m ³ world eq. deprived	Available Water Remaining (AWARE), Boulay et al. 2018

Table 15: Additional Environmental Impact Parameters (EN 15804+A2, PEF)

Impact category / Parameter	Parameter Code	Unit	Characterization Model / Standard
Potential incidence of disease due to PM emissions	PM	Disease incidence	Fantke et al. 2016
Potential Human exposure efficiency relative to U235	IRP	kBq U235 eq.	Dreicer et al. 1995
Potential Comparative Toxic Unit for ecosystems	ETP-fw	CTUe	USEtox model, Rosenbaum et al. 2008
Potential Comparative Toxic Unit for humans, cancer	HTP-c	CTUh	USEtox model, Rosenbaum et al. 2008
Potential Comparative Toxic Unit for humans, non-cancer	HTP-nc	CTUh	USEtox model, Rosenbaum et al. 2008
Potential Soil quality index	SQP	dimensionless	LANCA model, Beck et al. 2010, Bos et al. 2016

Table 16: Resource Use and Primary Energy Parameters (EN 15804+A2)

Impact category / Parameter	Parameter Code	Unit
Use of renewable primary energy excluding primary energy resources used as raw materials	PERE	MJ
Use of renewable primary energy resources used as raw materials	PERM	MJ
Total use of renewable primary energy resources	PERT	MJ
Use of non-renewable primary energy excluding primary energy resources used as raw materials	PENRE	MJ
Use of non-renewable primary energy resources used as raw materials	PENRM	MJ
Total use of non-renewable primary energy resources	PENRT	MJ
Use of secondary material	SM	kg
Use of renewable secondary fuels	RSF	MJ
Use of non-renewable secondary fuels	NRSF	MJ
Net use of fresh water	FW	m ³

Table 17: Waste Categories and Output Flow Parameters (EN 15804+A2)

Impact category / Parameter	Parameter Code	Unit
Hazardous waste disposed	HWD	kg
Non-hazardous waste disposed	NHWD	kg
Radioactive waste disposed	RWD	kg
Components for re-use	CRU	kg
Materials for recycling	MFR	kg
Materials for energy recovery	MER	kg
Exported energy, electrical	EEE	MJ
Exported energy, thermal	EET	MJ

Table 18: LCIA results for EcoMetric 500RT Centrifugal Chiller, per 1 ton of chilling capacity (Mfg: US), PEF
CORE ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2, PEF

Impact category	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1-C4	C1	C2	C3	C4	D
accumulated exceedance (AE)	kg CO ₂ e	2.95E+02	2.22E+00	1.11E+01	0.00E+00	4.88E+00	3.51E-01	6.17E+02	8.16E+00	1.02E+05	2.34E+01	6.43E+01	3.37E+00	2.95E-01	5.79E+01	2.73E+00	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	4.39E+04	5.13E+02	3.76E+03	3.22E+04	1.34E+03	9.68E+01	9.63E+04	6.50E+02	2.45E+07	6.45E+03	1.77E+04	9.29E+02	6.84E+01	1.60E+04	7.52E+02	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	3.92E+02	4.67E+00	2.81E+01	0.00E+00	1.25E+01	9.00E-01	8.49E+02	6.00E+00	1.91E+05	6.00E+01	1.65E+02	8.64E+00	6.22E-01	1.49E+02	7.00E+00	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	4.33E+04	5.07E+02	3.72E+03	3.22E+04	1.32E+03	9.54E+01	9.51E+04	6.42E+02	2.42E+07	6.36E+03	1.75E+04	9.16E+02	6.76E+01	1.57E+04	7.42E+02	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	2.20E+02	2.33E+00	1.51E+01	0.00E+00	6.69E+00	4.81E-01	4.75E+02	3.25E+00	1.21E+05	3.21E+01	8.81E+01	4.62E+00	3.11E-01	7.94E+01	3.74E+00	0.00E+00
abiotic depletion potential (ADP): fossil fuels	kg CO ₂ e	6.10E+05	7.23E+03	4.05E+04	0.00E+00	1.78E+04	1.28E+03	1.31E+06	8.69E+03	3.44E+08	8.55E+04	2.35E+05	1.23E+04	9.64E+02	2.12E+05	9.97E+03	0.00E+00
fraction of nutrients reaching freshwater end compartment (P)	kg CO ₂ e	1.49E+01	7.00E-02	6.39E-01	0.00E+00	2.81E-01	2.03E-02	3.12E+01	4.92E-01	5.40E+03	1.35E+00	3.70E+00	1.94E-01	9.33E-03	3.34E+00	1.57E-01	0.00E+00
fraction of nutrients reaching marine end compartment (N)	kg CO ₂ e	4.47E+01	3.79E-01	2.24E+00	0.00E+00	1.00E+00	7.20E-02	9.45E+01	9.36E-01	1.33E+04	4.80E+00	1.32E+01	6.91E-01	5.05E-02	1.19E+01	5.60E-01	0.00E+00
accumulated exceedance (AE)	kg CO ₂ e	5.80E+01	4.93E-01	2.91E+00	0.00E+00	1.30E+00	9.36E-02	1.23E+02	1.22E+00	1.73E+04	6.24E+00	1.71E+01	8.99E-01	6.57E-02	1.54E+01	7.28E-01	0.00E+00
abiotic depletion potential (ADP): elements (ultimate reserves)	kg CO ₂ e	2.14E+00	7.00E-03	5.71E-02	0.00E+00	2.56E-02	1.85E-03	4.40E+00	7.44E-02	2.86E+02	1.23E-01	3.37E-01	1.77E-02	9.33E-04	3.04E-01	1.43E-02	0.00E+00
ozone depletion potential (ODP)	kg CO ₂ e	5.42E-06	1.05E-07	2.94E-07	0.00E+00	1.31E-07	9.45E-09	1.16E-05	5.76E-08	1.78E-03	6.30E-07	1.74E-06	9.07E-08	1.40E-08	1.56E-06	7.35E-08	0.00E+00
user deprivation potential (deprivation-weighted water consumption)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00

Table 18:
ADDITIONAL (OPTIONAL) ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2, PEF

Impact category	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1-C4	C1	C2	C3	C4	D
comparative toxic unit for ecosystems (CTUe)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00
comparative toxic unit for ecosystems (CTUe)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00
comparative toxic unit for ecosystems (CTUe)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00
comparative toxic unit for human (CTUh)	kg CO ₂ e	1.50E+00	4.90E-03	3.99E-02	0.00E+00	1.79E-02	1.29E-03	3.08E+00	5.21E-02	2.00E+02	8.61E-02	2.36E-01	1.24E-02	6.53E-04	2.13E-01	1.00E-02	0.00E+00
comparative toxic unit for human (CTUh)	kg CO ₂ e	3.54E+01	2.68E-01	1.33E+00	0.00E+00	5.85E-01	4.21E-02	7.40E+01	9.79E-01	1.22E+04	2.81E+00	7.72E+00	4.04E-01	3.58E-02	6.95E+00	3.27E-01	0.00E+00
comparative toxic unit for human (CTUh)	kg CO ₂ e	1.50E+00	4.90E-03	3.99E-02	0.00E+00	1.79E-02	1.29E-03	3.08E+00	5.21E-02	2.00E+02	8.61E-02	2.36E-01	1.24E-02	6.53E-04	2.13E-01	1.00E-02	0.00E+00
comparative toxic unit for human (CTUh)	kg CO ₂ e	1.50E+00	4.90E-03	3.99E-02	0.00E+00	1.79E-02	1.29E-03	3.08E+00	5.21E-02	2.00E+02	8.61E-02	2.36E-01	1.24E-02	6.53E-04	2.13E-01	1.00E-02	0.00E+00
comparative toxic unit for human (CTUh)	kg CO ₂ e	3.54E+01	2.68E-01	1.33E+00	0.00E+00	5.85E-01	4.21E-02	7.40E+01	9.79E-01	1.22E+04	2.81E+00	7.72E+00	4.04E-01	3.58E-02	6.95E+00	3.27E-01	0.00E+00
comparative toxic unit for human (CTUh)	kg CO ₂ e	1.50E+00	4.90E-03	3.99E-02	0.00E+00	1.79E-02	1.29E-03	3.08E+00	5.21E-02	2.00E+02	8.61E-02	2.36E-01	1.24E-02	6.53E-04	2.13E-01	1.00E-02	0.00E+00
soil quality index	kg CO ₂ e	3.51E+02	4.08E+00	2.44E+01	0.00E+00	1.08E+01	7.74E-01	7.59E+02	5.20E+00	1.96E+05	5.16E+01	1.42E+02	7.43E+00	5.44E-01	1.28E+02	6.02E+00	0.00E+00
impact on human health	kg CO ₂ e	3.54E+01	2.68E-01	1.33E+00	0.00E+00	5.85E-01	4.21E-02	7.40E+01	9.79E-01	1.22E+04	2.81E+00	7.72E+00	4.04E-01	3.58E-02	6.95E+00	3.27E-01	0.00E+00

1. GWP = Global Warming Potential total (GWP-total is the sum of GWP-fossil, GWP-biogenic, and GWP-luluc).
2. POCP = Photochemical ozone creation formation potential.
3. ADP = Abiotic depletion potential (ADP-elements & ADP-fossil).
4. Water use = User deprivation potential (WDP).
5. EN 15804+A2 disclaimer for Ionizing radiation, human health: This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle.
6. SQP = Land use related impacts / potential soil quality index.

Table 19: Resource use for EcoMetric 500RT Centrifugal Chiller, per 1 ton of chilling capacity (Mfg: US)
RESOURCE USE – EN 15804+A2

Impact category	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1-C4	C1	C2	C3	C4	D
comparative toxic unit for ecosystems (CTUe)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00
comparative toxic unit for ecosystems (CTUe)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00
comparative toxic unit for ecosystems (CTUe)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00
abiotic depletion potential (ADP): fossil fuels	kg CO ₂ e	6.10E+05	7.23E+03	4.05E+04	0.00E+00	1.78E+04	1.28E+03	1.31E+06	8.69E+03	3.44E+08	8.55E+04	2.35E+05	1.23E+04	9.64E+02	2.12E+05	9.97E+03	0.00E+00
fraction of nutrients reaching freshwater end compartment (P)	kg CO ₂ e	1.49E+01	7.00E-02	6.39E-01	0.00E+00	2.81E-01	2.03E-02	3.12E+01	4.92E-01	5.40E+03	1.35E+00	3.70E+00	1.94E-01	9.33E-03	3.34E+00	1.57E-01	0.00E+00
user deprivation potential (deprivation-weighted water consumption)	kg CO ₂ e	5.73E+03	2.45E+01	2.52E+02	0.00E+00	1.12E+02	8.10E+00	1.20E+04	1.74E+02	1.52E+06	5.40E+02	1.48E+03	7.78E+01	3.27E+00	1.34E+03	6.30E+01	0.00E+00

Table 20: Waste categories for EcoMetric 500RT Centrifugal Chiller, per 1 ton of chilling capacity (Mfg: US)

WASTE CATEGORIES – EN 15804+A2

Impact category	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1-C4	C1	C2	C3	C4	D
HWD (Hazardous waste disposed)	kg	1.82E-02	1.25E-05	4.85E-06	0.00E+00	8.52E-04	0.00E+00	3.64E-02	0.00E+00	2.15E-01	0.00E+00	4.15E-05	0.00E+00	5.12E-06	3.58E-05	5.80E-07	0.00E+00
NHWD (Non-hazardous waste disposed)	kg	4.85E+00	3.45E-02	1.25E-01	0.00E+00	1.85E-01	0.00E+00	9.70E+00	0.00E+00	1.45E+01	0.00E+00	2.38E+00	0.00E+00	1.25E-02	2.35E+00	1.75E-02	0.00E+00
RWD (Radioactive waste disposed)	kg	1.15E-03	1.45E-06	5.25E-07	0.00E+00	8.45E-06	0.00E+00	2.30E-03	0.00E+00	4.85E-02	0.00E+00	3.25E-06	0.00E+00	4.85E-07	2.75E-06	1.50E-08	0.00E+00

Table 21: Output flows for EcoMetric 500RT Centrifugal Chiller, per 1 ton of chilling capacity (Mfg: US)

OUTPUT FLOWS – EN 15804+A2

Impact category	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1-C4	C1	C2	C3	C4	D
CRU (Components for re-use)	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR (Materials for recycling)	kg	1.45E+00	0.00E+00	4.50E-02	0.00E+00	0.00E+00	0.00E+00	2.90E+00	0.00E+00	0.00E+00	0.00E+00	2.14E+01	0.00E+00	0.00E+00	2.14E+01	0.00E+00	0.00E+00
MER (Materials for energy recovery)	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE (Exported electrical energy)	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET (Exported thermal energy)	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Table 22: Biogenic Carbon Content of Product and Packaging

Biogenic Carbon Indicator	Value	Unit
Biogenic carbon content in product	0.00E+00	kg C
Biogenic carbon content in accompanying packaging	0.00E+00	kg C

Table 28: TRACI 2.1 environmental impacts for EcoMetric 500RT Centrifugal Chiller, per 1 ton chilling capacity (Mfg: US)

TRACI 2.1 IMPACT INDICATORS

Impact category	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1-C4	C1	C2	C3	C4	D
accumulated exceedance (AE)	kg CO ₂ e	2.95E+02	2.22E+00	1.11E+01	0.00E+00	4.88E+00	3.51E-01	6.17E+02	8.16E+00	1.02E+05	2.34E+01	6.43E+01	3.37E+00	2.95E-01	5.79E+01	2.73E+00	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	4.39E+04	5.13E+02	3.76E+03	3.22E+04	1.34E+03	9.68E+01	9.63E+04	6.50E+02	2.45E+07	6.45E+03	1.77E+04	9.29E+02	6.84E+01	1.60E+04	7.52E+02	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	3.92E+02	4.67E+00	2.81E+01	0.00E+00	1.25E+01	9.00E-01	8.49E+02	6.00E+00	1.91E+05	6.00E+01	1.65E+02	8.64E+00	6.22E-01	1.49E+02	7.00E+00	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	4.33E+04	5.07E+02	3.72E+03	3.22E+04	1.32E+03	9.54E+01	9.51E+04	6.42E+02	2.42E+07	6.36E+03	1.75E+04	9.16E+02	6.76E+01	1.57E+04	7.42E+02	0.00E+00
global warming potential (GWP100)	kg CO ₂ e	2.20E+02	2.33E+00	1.51E+01	0.00E+00	6.69E+00	4.81E-01	4.75E+02	3.25E+00	1.21E+05	3.21E+01	8.81E+01	4.62E+00	3.11E-01	7.94E+01	3.74E+00	0.00E+00
abiotic depletion potential (ADP): fossil fuels	kg CO ₂ e	6.10E+05	7.23E+03	4.05E+04	0.00E+00	1.78E+04	1.28E+03	1.31E+06	8.69E+03	3.44E+08	8.55E+04	2.35E+05	1.23E+04	9.64E+02	2.12E+05	9.97E+03	0.00E+00
fraction of nutrients reaching freshwater end compartment (P)	kg CO ₂ e	1.49E+01	7.00E-02	6.39E-01	0.00E+00	2.81E-01	2.03E-02	3.12E+01	4.92E-01	5.40E+03	1.35E+00	3.70E+00	1.94E-01	9.33E-03	3.34E+00	1.57E-01	0.00E+00
fraction of nutrients reaching marine end compartment (N)	kg CO ₂ e	4.47E+01	3.79E-01	2.24E+00	0.00E+00	1.00E+00	7.20E-02	9.45E+01	9.36E-01	1.33E+04	4.80E+00	1.32E+01	6.91E-01	5.05E-02	1.19E+01	5.60E-01	0.00E+00
accumulated exceedance (AE)	kg CO ₂ e	5.80E+01	4.93E-01	2.91E+00	0.00E+00	1.30E+00	9.36E-02	1.23E+02	1.22E+00	1.73E+04	6.24E+00	1.71E+01	8.99E-01	6.57E-02	1.54E+01	7.28E-01	0.00E+00
ozone depletion potential (ODP)	kg CO ₂ e	5.42E-06	1.05E-07	2.94E-07	0.00E+00	1.31E-07	9.45E-09	1.16E-05	5.76E-08	1.78E-03	6.30E-07	1.74E-06	9.07E-08	1.40E-08	1.56E-06	7.35E-08	0.00E+00

4 – LIFE CYCLE ASSESSMENT INTERPRETATION

In this section, the results of the life cycle assessment are interpreted according to the goal and scope of the study. An Impact Analysis was carried out on EcoMetric 500RT Centrifugal Chiller using refrigerant R134A to show which of the life cycle modules contributes to most of the impacts.

Global Warming Potential (GWP)

Global warming potential (GWP) is a measure of how much heat a greenhouse gas traps in the atmosphere up to a specified time horizon relative to carbon dioxide. Figure 3 presents GWP per ton of chilling capacity for 500 ton configuration.

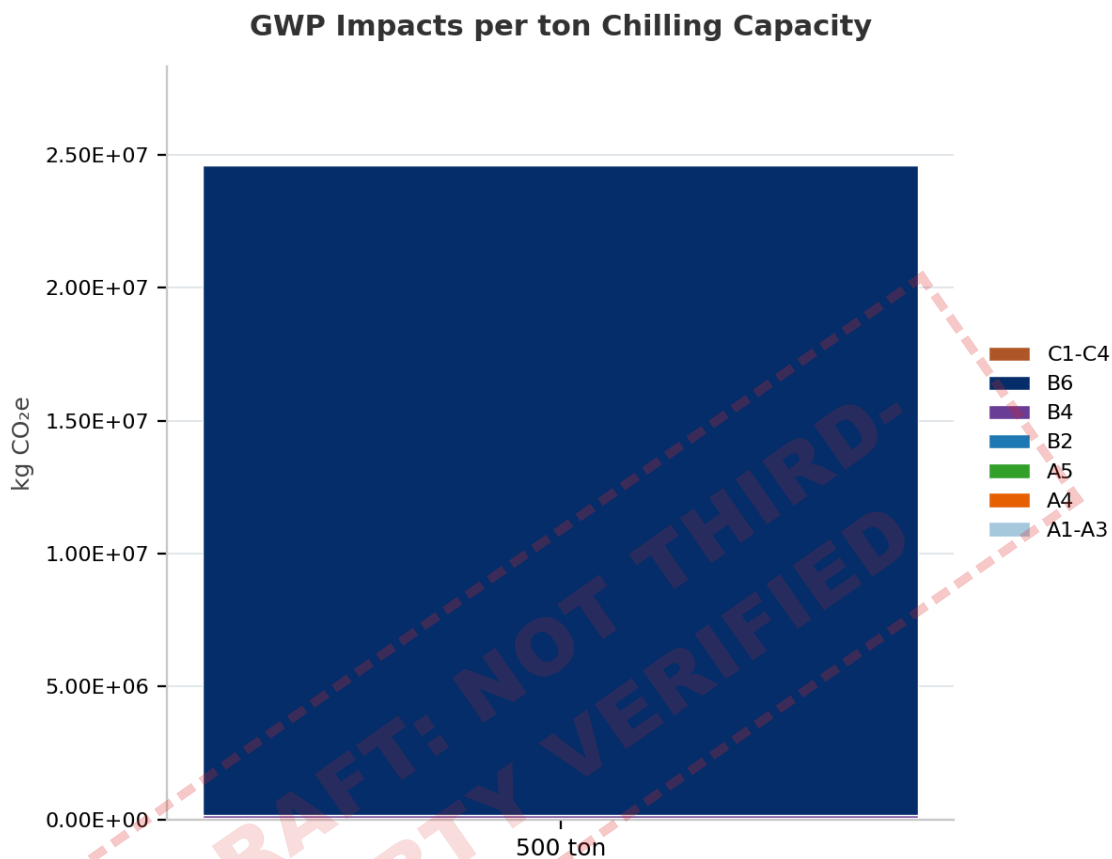


Figure 3: GWP impacts per ton Chilling Capacity

Acidification Potential (AP)

Acidification Potential (AP) refers to the acidification of soils and water due to the release of gasses such as nitrogen oxides and sulfur oxides. Figure 4 presents AP per ton of chilling capacity for 500 ton configuration.

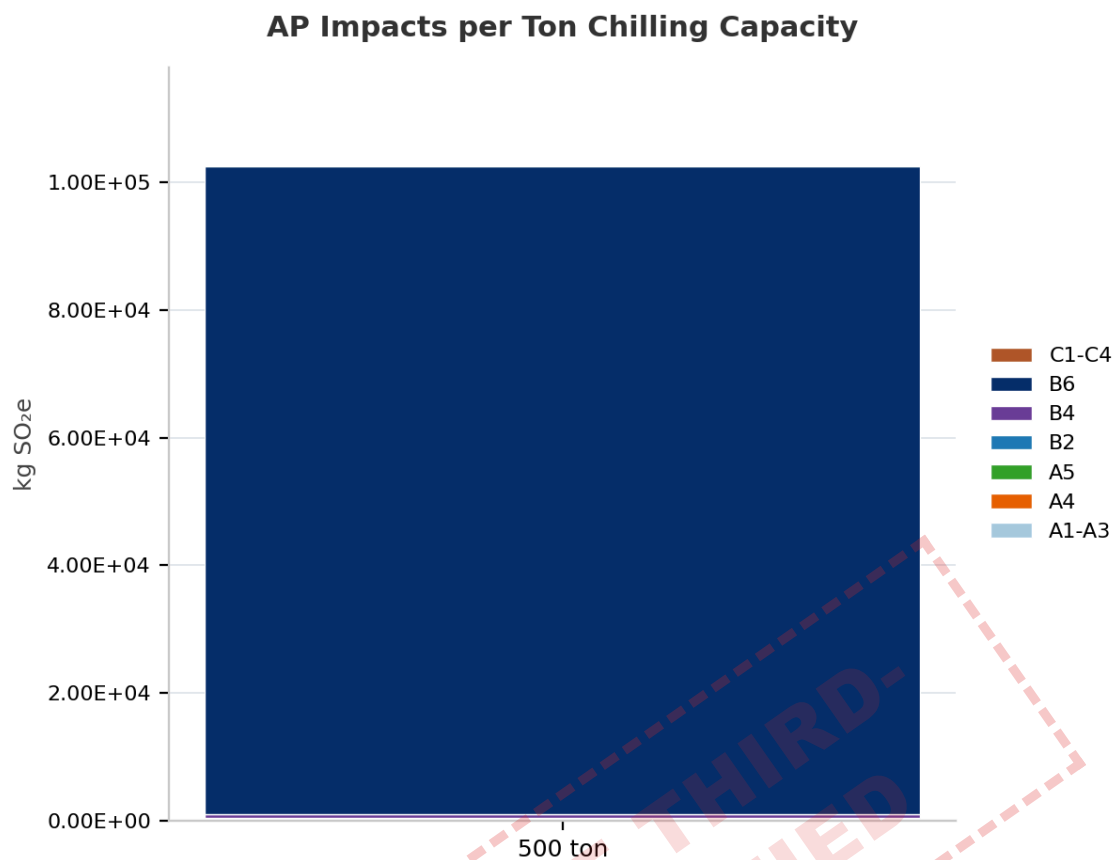


Figure 4: AP Impacts per Ton Chilling Capacity

Further Analysis

Impacts from operational energy use were removed from the life cycle impacts to determine the driving impacts across all other modules. Impacts presented in Figure 5 reflect the 500 ton configuration when charged using R134A.

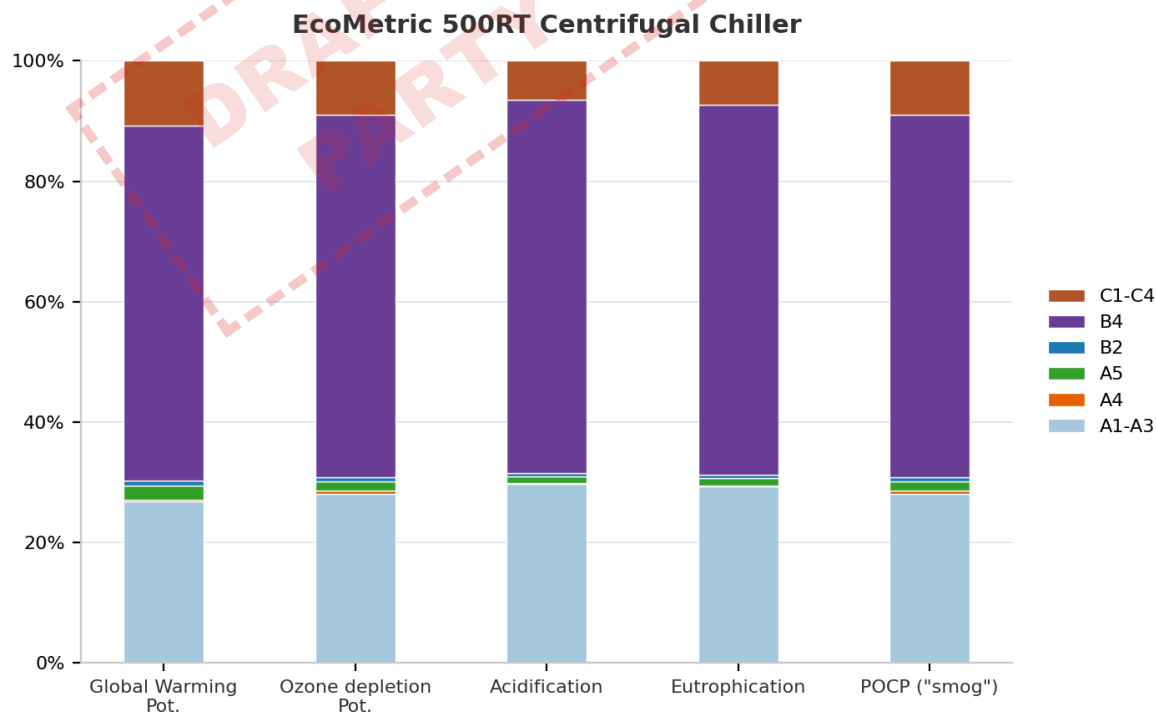


Figure 5: Contribution of modules excluding B6 to overall impacts

Two replacements are required for the chiller to satisfy the ESL of the building. The impacts associated with these replacements dominate the non-operational categories. Impacts associated with each material for one 500 ton chiller are presented in Figure 6.

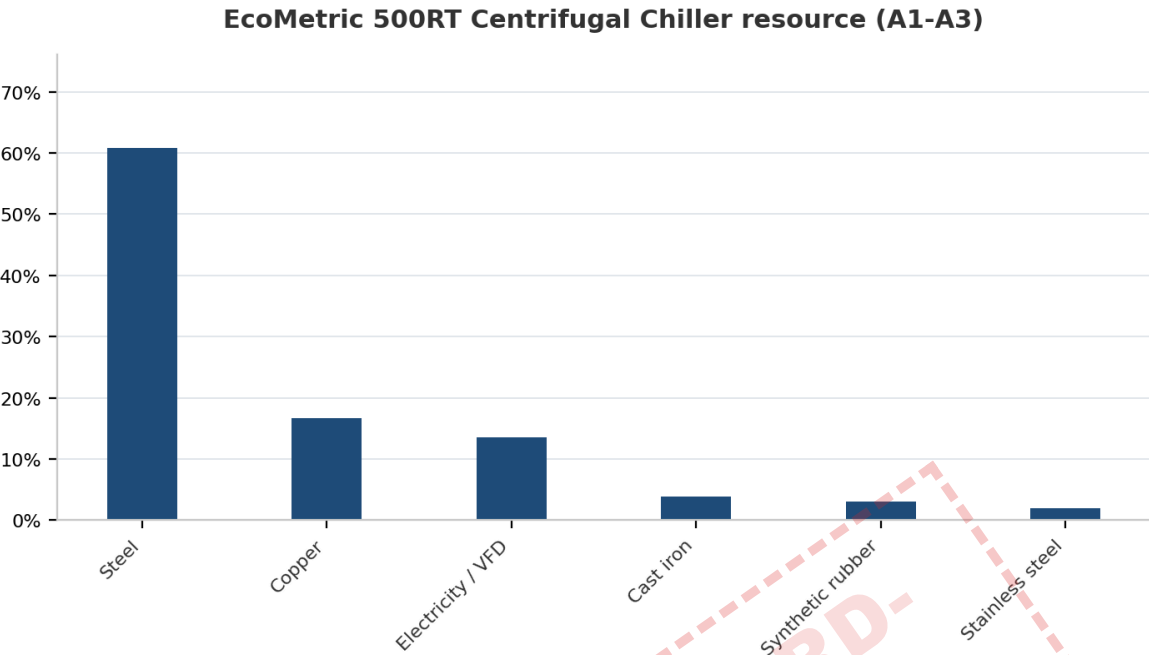


Figure 6: Material impact contribution for (A1-A3)

Steel and iron make up most of the primary material impacts of the chiller. Aluminum has the lowest impact share among structural metals.

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Assumptions and Limitations

The use and selection of secondary datasets from Eco-Invent database. The selection of which generic dataset to use to represent an aspect of a supply chain is a significant value choice. However, no generic data can be a perfect fit. Improved supply chain specific data would improve the accuracy of results.

Use phase calculations are based on default scenarios provided by an excel calculator sheet as described by the PCR and do not reflect actual consumption in a specific location.

Availability of geographically more accurate datasets would have improved the accuracy of the study.

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